

Missing Coin Sum

Time limit: 1.00 s

Memory limit: 512 MB

You have n coins with positive integer values. What is the smallest sum you cannot create using a subset of the coins?

Input

The first line has an integer n : the number of coins.

The second line has n integers $x_1, x_2, \dots, x_n$: the value of each coin.

Output

Print one integer: the smallest coin sum.

Constraints

- $1 \leq n \leq 2 \cdot 10^5$
- $1 \leq x_i \leq 10^9$

Example

Input	Output
5 2 9 1 2 7	6